Template-free optimal pulse processing for saturating microcalorimeters: shift-invariant whitened PCA manifolds and Woodbury-accelerated non-stationary maximum-likelihood estimation

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The optimal filter used for microcalorimeter pulses assumes a fixed template, stationary noise and a known arrival time. Saturating transition-edge sensors (TES) violate all three. We present a template-free estimator in three stages. (A) Shift-invariant PCA in the noise-whitened domain learns a pulse subspace and sub-sample arrival times by classification EM; it is exact because spectral whitening commutes with Fourier shifts. (B) A spline manifold through the origin, parameterised by whitened arc length (the cumulative matched-filter SNR, with unit Fisher information), maps the coefficients to energy. (C) An energy-dependent, low-rank covariance $R(E, t_0) = R_0 + T_{t_0} G\Sigma(E) G^{\mathsf{T}} T_{t_0}^{\mathsf{T}}$ captures pulse-correlated noise and shape fluctuations. The Woodbury identity and the determinant lemma reduce the joint energy–time likelihood to about 12 dimensions. We prove that shape fluctuations along the manifold tangent cannot be learned from continuum data, but can be learned from mono-energetic clusters without knowing their energies. On a physical simulation of a saturating TES, tested on held-out lines, the estimator reaches 1.7–2.6 eV FWHM from 1 to 11 keV, against 2.0–12.4 eV for a plain optimal filter. Most of that gap is also closed by tuned empirical corrections. Against the best of four prior-art pipelines, the estimator is level at low energy and up to 18% better at 11 keV, within 0.04 eV of an oracle. On real single-line NIST TES data it matches, but does not beat, NIST's own MASS pipeline (0.263% against 0.254% FWHM). In a simulated phonon-mediated detector with line-free science data, covariances learned on a calibration run give 1.2–3.6 times better resolution than an energy-dependent full-covariance filter, whereas nothing learned from the science data alone removes the position dependence. A label-free sampling-phase correction brings arrival times close to the Cramér–Rao bound, but it needs about 10^5 records per energy. An open-source Python implementation accompanies the paper.

I. INTRODUCTION

Thermal microcalorimeters such as transition-edge sensors (TES), metallic magnetic calorimeters and thermal kinetic-inductance detectors reach energy resolutions of a few eV at keV energies [1, 2]. The standard estimator is the optimal filter (OF) [3, 4]: a frequency-domain matched filter that is the minimum-variance unbiased amplitude estimator if (i) every pulse is a scaled copy of one template, (ii) the noise is additive, Gaussian and stationary, and (iii) the arrival time is known. Real detectors break each assumption. As a TES is driven up its transition, the pulse shape evolves with energy and eventually saturates. The Johnson and thermal-fluctuation noise depend on the instantaneous resistance and temperature, so the noise during a pulse differs from the baseline noise. Arrival times are continuous, which leads to sampling-phase ("arrival-time") effects [4]. Position-dependent thermalisation adds shape variation that is not a function of energy.

Three lines of work address these effects. *Calibration-based non-linear and non-stationary filters* estimate energy-dependent templates and full covariance matrices at calibration energies and interpolate between them [5–9]. Fixsen *et al.* [8] gave this a differential-geometric interpretation, and Fowler *et al.* [10] showed that filtering along the *tangent* of the energy curve of mean pulses improves on the OF. These methods are the most accurate known. Their full-rank covariances, however, cost $\mathcal{O}(N^3)$ per energy and need many calibration pulses, a burden that led the Athena X-IFU study to judge them impractical [9]. *Empirical corrections* applied to the OF output are the practitioner's standard. They include arrival-phase and drift corrections [4] and decorrelation against residual "projector" components, as implemented in NIST's MASS software [11]. In dark-matter detectors, pulse-shape variation is treated as non-stationary noise added to the OF covariance [12, 13]. *Data-driven subspace methods* use principal component analysis (PCA) or singular value decomposition (SVD) of the records [14–21] and, more recently, neural networks [13, 22]. In earlier work [21] we proposed a template-free estimator based on a weighted EM-PCA [23, 24] of a phase-differenced Fourier transform that is invariant to arrival time. That construction discards arrival-time information by design, and its pulse-height estimator, the norm of the coefficient vector, is rectified at low signal-to-noise ratio.

This paper combines the three lines into one likelihood. It learns the pulse family from unlabelled data, as the subspace methods do. It models energy-dependent, non-stationary noise and shape fluctuations, as the calibration-based filters do, but with a low-rank parameterisation that is cheap to evaluate and needs few calibration pulses. It replaces the empirical corrections with a joint energy–time likelihood. Section II positions each component against prior work. The contributions are:

1. **Shift-invariant whitened PCA** (section IV). We

show that the symmetric spectral whitening operator $R_0^{-1/2}$ commutes with fractional Fourier shifts, whereas a Cholesky factor of the Toeplitz covariance does not (lemma 1). In the whitened domain, alignment (E-step) and subspace learning (M-step) of a transformation-invariant EM [25] become exact operations at sub-sample resolution. We prove monotone convergence (proposition 2) and show that the noise does not bias the learned subspace (proposition 1). We also prove that a subspace containing the time derivative of the pulse carries no timing information (proposition 3).

2. **Arc-length spline manifold** (section V). A data-driven realisation of the energy curve of Refs. [8, 10, 18]: a P-spline graph through the origin in whitened coefficient space, parameterised by whitened arc length. In that parameter the Fisher information is one, and the estimate is unbiased at zero energy (proposition 4).

3. **Low-rank, Woodbury-accelerated non-stationary ML** (section VI). We use an energy-dependent covariance on the signal subspace plus a few learned residual modes. With it, the exact non-stationary log-likelihood reduces to $K_b \ll N$ dimensions (theorem 1), as for low-rank Gaussian processes in pulsar timing [26, 27]. We prove an identifiability result (proposition 5): fluctuations along the manifold tangent cannot be learned from unlabelled continuum events, but can be learned from mono-energetic clusters, without knowing their energies.

4. **A physical simulation study** (sections VII and VIII) of a saturating TES, with held-out test lines. We compare against the OF, against OF with phase correction and residual-projector decorrelation (the practitioner's standard), against a stationary secant (tangent) filter, against an energy-dependent full-covariance filter, and against an oracle Gaussian model.

5. **Open-source software**: a tested Python package with a scikit-learn-like interface (appendix A).

II. RELATED WORK

a. Shift-invariant subspace learning. Jointly inferring per-sample shifts and learning components by EM is established in machine learning, as transformation-invariant clustering and factor analysis with FFT-computed shifts [25, 28]. In neuroimaging it appears as shift-invariant decompositions with frequency-domain delays [29]. In functional data analysis it is registration combined with PCA [30], and in cryo-electron microscopy it is multi-reference alignment [31]. Among microcalorimeter methods, Ref. [21] linearised the shift in a phase-differenced Fourier domain instead of estimating it. NIST's MASS builds SVD "projector" bases in which the arrival time enters as a derivative component [4, 11, 19]. Alpert *et al.* [17] built an SVD single-pulse model with sub-sample arrival times for pile-up rejection. We found no microcalorimeter method that alternates sub-sample alignment with re-learning a noise-whitened basis. Our additions to the generic algorithm are the whitening that makes it exact (lemma 1), continuous shifts and the timing degeneracy of proposition 3.

b. Energy curves and manifolds. The pulse family of a non-linear detector is a curve in record space [8]. Tangent filtering uses its local tangent from energy-specific mean pulses [10]. Fowler *et al.* [18] fitted cubic splines to linear-subspace components as functions of an energy estimator. Yan *et al.* [15] and Miller *et al.* [20] mapped PCA coefficients to energy with the help of calibration lines. Our curve is learned without labels as a graph through the origin in *whitened* coefficient space. Its whitened arc length equals the cumulative matched-filter SNR, which gives the parameter a statistical meaning (unit Fisher information) and makes the low-energy estimate unbiased.

c. Non-stationary noise. Energy-dependent full covariances in a least-squares fit go back to Fixsen *et al.* [5, 6], with simulated covariances in Ref. [7] and calibration-line covariances interpolated in energy in Athena's SIRENA software [9]. SuperCDMS add the covariance of template residuals, aligned to the arrival time, to the noise [12], and treat the shape uncertainty of PCA templates as non-stationary noise [13]. These covariances are full-rank, or their modes are fitted as free amplitudes. A white-plus-low-rank covariance evaluated with the Woodbury identity and the determinant lemma is standard for Gaussian-process noise in pulsar timing [26]. Lentati *et al.* [27] fit arrival times jointly with stochastic profile variations, and Moore and Gair [32] marginalise waveform-model error as a covariance in gravitational-wave inference. To our knowledge, a learned, low-rank, continuous $\Sigma(E)$ inside a joint energy–time likelihood has not been used for cryogenic detectors. Nor have we found the identifiability statement of proposition 5, although its geometry is that of Refs. [8, 10].

d. Software. The most complete open practitioner pipelines are NIST's MASS (whitened OF, projectors, phase and drift corrections) [11] and SuperCDMS's QETpy (OF variants, including non-linear and multi-template filters) [33]. SIRENA implements the covariance-matrix methods of Ref. [9]. None implements the alternating shift-invariant learning, the whitened arc-length manifold or the low-rank joint likelihood. A detailed survey is included with the code (`paper/PRIOR_ART.md`).

III. SIGNAL MODEL AND NOTATION

A record $\boldsymbol{y} \in \mathbb{R}^N$ sampled at rate f_s is modelled as

$$\boldsymbol{y} = T_{t_0}\, \boldsymbol{m}(E, \eta) + \boldsymbol{n}, \tag{1}$$

where $\boldsymbol{m}(E, \eta)$ is the noise-free pulse for deposited energy E and nuisance morphology η (e.g. absorption position), t_0 is the arrival time in samples, and $\boldsymbol{n}$ is zero-mean Gaussian noise. Let F be the unitary DFT. The (band-limited, circular) shift operator is

$$T_\tau = F^H \operatorname{diag}\!\left(e^{-2\pi i k \tau / N}\right) F, \qquad (T_\tau \boldsymbol{x})(t) = x(t - \tau), \tag{2}$$

with the Nyquist bin removed so that T_τ maps real signals to real signals and is orthogonal on the remaining band for any real τ. Its generator is the spectral derivative $D = F^H \operatorname{diag}(2\pi i k/N) F$, with $T_\tau = e^{-\tau D}$ and $D^\mathsf{T} = -D$. Baseline noise is approximated by a circulant covariance $R_0 = F^H \Lambda F$, with $\Lambda = \operatorname{diag}(\lambda_k)$ estimated from pulse-free records as $\lambda_k = \mathbb{E}|Y_k|^2/N$. This approximation is asymptotically equivalent to the Toeplitz covariance [34] and is standard in OF practice [3]. We exclude the DC bin (baseline offset) and the Nyquist bin, which leaves $N - 2$ real degrees of freedom.

Lemma 1 (Whitening commutes with shifts)**.** *Let $W = R_0^{-1/2} = F^H \Lambda^{-1/2} F$. Then $W T_\tau = T_\tau W$ for all $\tau \in \mathbb{R}$, and $W\boldsymbol{n}$ is white with unit variance on the analysis band. If G is any other whitening matrix ($G R_0 G^\mathsf{T} = I$), then $G = QW$ with Q orthogonal, and G commutes with all T_τ if and only if Q is diagonalised by F. In particular, the inverse Cholesky factor L_0^{-1} of a non-circulant Toeplitz R_0 does not commute with shifts.*

Proof. W and T_τ are both diagonal in the Fourier basis. For the second statement, $(GW^{-1})(GW^{-1})^\mathsf{T} = G R_0 G^\mathsf{T} = I$, so $Q = GW^{-1}$ is orthogonal, and Q commutes with the full family $\{T_\tau\}$ iff it is a function of D, i.e. diagonal in F. A lower-triangular L_0^{-1} is not circulant unless R_0 is diagonal. □

All subsequent processing works with whitened records $\boldsymbol{y}_\mathrm{w} = W\boldsymbol{y}$. A delay of the raw record is the same delay of the whitened record, so a shifted pulse remains a shifted pulse after whitening. This is the property that makes the methods below exact rather than approximate.

IV. STAGE A: SHIFT-INVARIANT WHITENED PCA

A. Model and objective

In the whitened domain we model the pulses of a training set as lying in an unknown K-dimensional subspace with orthonormal basis $B \in \mathbb{R}^{K \times N}$ ($BB^\mathsf{T} = I$):

$$\boldsymbol{y}_\mathrm{w}^{(j)} = T_{\tau_j} B^\mathsf{T} \boldsymbol{a}_j + \boldsymbol{\varepsilon}_j, \qquad \boldsymbol{\varepsilon}_j \sim \mathcal{N}(0, I). \tag{3}$$

The amplitudes $\boldsymbol{a}_j$ profile out as $\hat{\boldsymbol{a}}_j = B T_{-\tau_j} \boldsymbol{y}_\mathrm{w}^{(j)}$. The negative log-likelihood is then $\sum_j \|\boldsymbol{y}_\mathrm{w}^{(j)}\|^2 - J(B, \{\tau_j\})$ with captured power

$$J(B, \{\tau_j\}) = \sum_j \left\| B\, T_{-\tau_j} \boldsymbol{y}_\mathrm{w}^{(j)} \right\|^2. \tag{4}$$

We maximise J by alternating over the latent shifts and the basis. This is a hard (classification) EM, a shift-invariant analogue of EM-PCA [23, 35].

B. E-step: phase-coherent cross-correlation

For fixed B, each τ_j maximises $g_j(\tau) = \|\boldsymbol{x}_j(\tau)\|^2$, where $x_{jk}(\tau) = \langle \boldsymbol{b}_k, T_{-\tau} \boldsymbol{y}_\mathrm{w}^{(j)} \rangle$. Writing $\hat{\cdot}$ for the real FFT and $w_k \in \{1, 2\}$ for the multiplicity of each half-spectrum bin,

$$x_{jk}(\tau) = \frac{1}{N} \sum_{k'} w_{k'}\, \Re\!\left[\overline{\hat{b}_{k,k'}}\, \hat{y}_{j,k'}\, e^{2\pi i k' \tau / N} \right]. \tag{5}$$

For all integer lags this is a cross-correlation, computed with one inverse FFT per basis vector. The maximum is then refined to sub-sample precision with guarded Newton steps on the exact band-limited function (5). Its τ-derivatives follow analytically by multiplying the summand by $(2\pi i k'/N)^m$, at $\mathcal{O}(KN)$ cost per step. Unlike up-sampling or parabolic interpolation, this optimises the true likelihood, so the estimate inherits the asymptotic efficiency of maximum likelihood.

C. M-step: counter-rotation and SVD

For fixed $\{\tau_j\}$, J is maximised by the top-K right singular vectors of the matrix X whose rows are the counter-rotated records $T_{-\tau_j} \boldsymbol{y}_\mathrm{w}^{(j)}$. The counter-rotation is a phase ramp in the frequency domain. We compute the $N \times N$ scatter matrix $X^\mathsf{T} X$ and its eigendecomposition.

Proposition 1 (Whitened noise does not bias the subspace)**.** *If $\boldsymbol{y}_\mathrm{w}^{(j)} = T_{\tau_j} \boldsymbol{\mu}_j + \boldsymbol{\varepsilon}_j$ with $\boldsymbol{\varepsilon}_j$ white and the shifts are known, then $\mathbb{E}[X^\mathsf{T} X]/n = \frac{1}{n} \sum_j \boldsymbol{\mu}_j \boldsymbol{\mu}_j^\mathsf{T} + P$, where P projects onto the analysis band. The noise term is isotropic, so the eigenvectors are those of the noise-free scatter matrix.*

Proof. $T_{-\tau_j}$ is orthogonal and commutes with P, so $\mathbb{E}[T_{-\tau_j} \boldsymbol{\varepsilon}_j \boldsymbol{\varepsilon}_j^\mathsf{T} T_{-\tau_j}^\mathsf{T}] = P$. □

The proposition fails without whitening: coloured noise adds a non-isotropic term R_0 and tilts the subspace toward the noise-dominated frequencies. With *estimated* shifts, low-SNR records are aligned partly to their own noise. This is the "template from noise" bias known from cryo-electron microscopy [31, 36]. We therefore learn B only from records whose whitened excess power exceeds $\mathrm{SNR}^2_{\min}$. All records are still aligned and projected.

D. Convergence, gauge and a timing degeneracy

Proposition 2 (Monotonicity)**.** *Let the E-step maximise* $J_r = \sum_j \|B_{1:r} T_{-\tau_j} \boldsymbol{y}_{\mathrm{w}}^{(j)}\|^2$ *over* τ_j *for the leading* $r \le K$ *basis vectors, and let the M-step take the top-*K *singular vectors. Then* J_r *is non-decreasing over the iterations.*

Proof. The E-step maximises J_r for fixed B. The top-r right singular vectors maximise $\sum_j \|B_{1:r}\boldsymbol{x}_j\|^2$ over all r-frames (Ky Fan), and they are the first r of the top-K singular vectors. □

The model is invariant under a common shift $(B, \tau_j) \mapsto (T_\delta B, \tau_j + \delta)$. We fix this gauge by setting the median arrival time of the training set to zero after every iteration; this leaves J_r unchanged. The choice of r is governed by the following result.

Proposition 3 (A subspace containing the pulse derivative carries no timing information)**.** *Let* P_B *be the projector onto* $\mathrm{span}(B)$ *and* $g(\tau) = \|P_B T_{-\tau}\boldsymbol{y}\|^2$*. At a point where* $\boldsymbol{z} = T_{-\tau}\boldsymbol{y} \in \mathrm{span}(B)$*,*

$$g''(\tau) = -2\,\|(I - P_B)\,D\boldsymbol{z}\|^2. \tag{6}$$

The curvature of the profile likelihood in τ*, and so the Fisher information of the subspace estimator of* t_0*, comes only from the part of the time derivative of the pulse that lies* outside *the subspace.*

Proof. $\frac{d}{d\tau}\boldsymbol{z} = D\boldsymbol{z}$. Then $g' = 2\boldsymbol{z}^{\mathsf{T}} P_B D\boldsymbol{z}$ and $g'' = 2(D\boldsymbol{z})^{\mathsf{T}} P_B (D\boldsymbol{z}) + 2\boldsymbol{z}^{\mathsf{T}} P_B D^2 \boldsymbol{z}$. With $P_B\boldsymbol{z} = \boldsymbol{z}$ and $D^{\mathsf{T}} = -D$, the second term equals $-2\|D\boldsymbol{z}\|^2$, and $\|D\boldsymbol{z}\|^2 - \|P_B D\boldsymbol{z}\|^2 = \|(I - P_B)D\boldsymbol{z}\|^2$. □

A free K-dimensional subspace learned from jittered data tends to absorb the derivative direction, since it explains small shifts linearly, and its E-step then becomes ill-posed. We observed exactly this for $K \ge 3$ with $r = K$ (timing scatter of a full sample). With $r = 1$ the alignment uses only the leading, mean-pulse direction $\boldsymbol{b}_1$. Since $\langle \boldsymbol{b}_1, D\boldsymbol{b}_1\rangle = 0$, proposition 3 then gives the classical OF timing information $a_1^2\|D\boldsymbol{b}_1\|^2$. The remaining $K - 1$ directions describe shape variation *after* alignment. The final timing estimate comes from the manifold-constrained likelihood of Stage C, where the pulse family is one-dimensional and cannot absorb shifts.

V. STAGE B: ARC-LENGTH SPLINE MANIFOLD

A. Representation

The aligned coefficients $\boldsymbol{a}_j = B T_{-\tau_j}\boldsymbol{y}_{\mathrm{w}}^{(j)} \in \mathbb{R}^K$ have unit isotropic noise (proposition 1) and concentrate near a curve, the image of the energy axis. Let $\boldsymbol{e}$ be the leading uncentred principal direction of $\{\boldsymbol{a}_j\}$ (the mean-pulse direction), $u = \langle \boldsymbol{e}, \boldsymbol{a}\rangle$, and let the columns of $\tilde{W}$ span the orthogonal complement. We represent the curve as the graph

$$\boldsymbol{c}(u) = \boldsymbol{e}\,u + \tilde{W}\,u\,\boldsymbol{h}(u), \qquad \boldsymbol{h} = \text{cubic P-spline}, \tag{7}$$

fitted by penalised least squares [37]. The penalty is selected by generalised cross-validation [38]. Three properties are built in. The curve passes through the origin (a zero-energy event is no pulse) and is linear for small signals. It is continued linearly for $u < 0$ and tangentially beyond the data. Being a graph over u, it cannot fold back on itself. Free-form principal curves [39], which we tried first, did fold back in sparse, strongly saturated regions.

a. Mono-energetic clusters and the secant fallback. The scatter of records within a mono-energetic line is, by definition, not along the energy direction (proposition 5). Left in the fit, it can tilt the curve's tangent toward the dominant within-line fluctuation, which is typically a sampling-phase mode. After a first pass, we therefore refit the curve with each cluster found in the spectrum (section VI C) replaced by its mean, weighted by its size. If too few records lie outside the clusters to determine the curvature (we require 30 per spline coefficient), the curve is taken as the shape-preserving secant through the origin and the cluster means. This is the learned analogue of the secant or tangent filter [10], and the case of real data dominated by a single line (section VIII I).

B. Arc length and the Fisher information

We re-parameterise by whitened arc length, $s(u) = \int_0^u \|\boldsymbol{c}'(v)\|\,dv$, so that $\|d\boldsymbol{c}/ds\| = 1$. Under stationary noise with known alignment, $\boldsymbol{a} = \boldsymbol{c}(s) + \boldsymbol{\varepsilon}$ with $\boldsymbol{\varepsilon} \sim \mathcal{N}(0, I_K)$, the Fisher information is $I_{ss} = \|d\boldsymbol{c}/ds\|^2 = 1$, so

$$\sigma_s \ge 1, \qquad \sigma_E \ge \left|\frac{dE}{ds}\right|, \qquad \mathrm{FWHM}_E \ge 2.355\left|\frac{dE}{ds}\right|. \tag{8}$$

In the linear regime, $\boldsymbol{c}(s) = s\boldsymbol{e}$ and $s = \langle \boldsymbol{e}, \boldsymbol{a}\rangle$ is the OF amplitude divided by its baseline resolution, i.e. the matched-filter SNR. The arc length is therefore the natural generalisation of the OF SNR to evolving pulse shapes. The ML estimate given the alignment is the closest point of the curve.

When the arrival time is estimated jointly, $\partial_s\boldsymbol{\mu} = B^{\mathsf{T}}\boldsymbol{c}'$ and $\partial_\tau\boldsymbol{\mu} = -D\boldsymbol{\mu}$ with $\boldsymbol{\mu} = B^{\mathsf{T}}\boldsymbol{c}(s)$. In the linear regime $\langle \boldsymbol{c}', D\boldsymbol{\mu}\rangle \propto \langle \boldsymbol{b}_1, D\boldsymbol{b}_1\rangle = 0$ and energy and time are Fisher-orthogonal. When the shape evolves with energy they are not: the cross term is the information-geometric origin of time walk, and the CRB on s inflates by $1/(1 - \rho^2)$, where ρ is the Fisher correlation coefficient. The joint fit of Stage C accounts for this automatically.

C. No rectification at low energy

Proposition 4 (Rectification of the coefficient norm)**.** *If* $\boldsymbol{a} \sim \mathcal{N}(\boldsymbol{c}, I_K)$ *then* $\mathbb{E}\|\boldsymbol{a}\|^2 = \|\boldsymbol{c}\|^2 + K$ *and, for* $\boldsymbol{c} = 0$, $\|\boldsymbol{a}\|$ *follows a* χ_K *distribution with mean* $\sqrt{2}\,\Gamma(\frac{K+1}{2})/\Gamma(\frac{K}{2}) > 0$. *The manifold coordinate with linear continuation through the origin satisfies* $\hat{s} = \langle \boldsymbol{e}, \boldsymbol{a} \rangle \sim \mathcal{N}(0,1)$ *for* $\boldsymbol{c} = 0$*, to the extent that the curve is straight within a few noise units of the origin, and is unbiased.*

The norm estimator of Ref. [21] therefore has an energy-dependent bias of order $K/(2\|\boldsymbol{c}\|)$ that vanishes only at high SNR, plus a non-linear response near threshold. The manifold coordinate has neither.

D. Energy scale

The manifold, and hence s, is learned without labels. The map $E(s)$ is monotone and, in the linear regime, proportional, but its curvature at high energy cannot be recovered from pulse shapes alone. We fit it with a shape-preserving cubic interpolant through the origin and the positions of known lines, as for any non-linear detector. This is the sense in which the method is template-free but not calibration-free.

VI. STAGE C: WOODBURY-ACCELERATED NON-STATIONARY MAXIMUM LIKELIHOOD

A. Covariance model

Pulse-correlated noise has several sources: resistance- and temperature-dependent Johnson and phonon noise, residual shape fluctuations such as position dependence, and the shape distortion caused by sampling a pulse that is not band-limited at a random sub-sample phase. We model all of them by a low-rank, energy-dependent excess covariance. In the whitened frame aligned to the pulse,

$$\boldsymbol{y}_{\mathrm{w}} \sim \mathcal{N}\big(T_\tau \boldsymbol{\mu}(s),\; I + T_\tau V^{\mathsf{T}} \Sigma(s) V T_\tau^{\mathsf{T}}\big)\,, \tag{9}$$

with $V \in \mathbb{R}^{K_v \times N}$ orthonormal and $\Sigma(s)$ a symmetric $K_v \times K_v$ matrix with $\Sigma(0) = 0$ and eigenvalues > -1. We take $V = \left[\begin{smallmatrix} B \\ V_r \end{smallmatrix}\right]$: the signal basis plus K_n residual modes V_r orthogonal to it, so that $K_v = K + K_n$. Including B matters. The energy tangent $B^{\mathsf{T}}\boldsymbol{c}'(s)$ lies in $\mathrm{span}(B)$, and the in-pulse noise along these directions can differ from baseline in either sign. A model built only from the largest residual modes misses this (section VIII). In the raw domain, using lemma 1, this is

$$R(E, t_0) = R_0 + T_{t_0}\, G\, \Sigma(E)\, G^{\mathsf{T}}\, T_{t_0}^{\mathsf{T}}, \qquad G = R_0^{1/2} V^{\mathsf{T}}, \tag{10}$$

which is the form of the original proposal. Because T_{t_0} commutes with R_0, the Gram matrix $G^{\mathsf{T}} T_{t_0}^{\mathsf{T}} R_0^{-1} T_{t_0} G = VV^{\mathsf{T}} = I$ does not depend on t_0. This is why the shift does not spoil the low-rank algebra. Negative eigenvalues of Σ are allowed: in a saturating TES the noise during a pulse can be *lower* than the baseline noise (section VIII).

B. Reduction to K_b dimensions

Theorem 1. *Let* $\tilde{B} \in \mathbb{R}^{K_b \times N}$ *be an orthonormal basis of* $\mathrm{span}(B, V)$, $\boldsymbol{x}(\tau) = \tilde{B} T_{-\tau} \boldsymbol{y}_{\mathrm{w}}$, $\boldsymbol{m}(s) = \tilde{B} B^{\mathsf{T}} \boldsymbol{c}(s)$, $Q = \tilde{B} V^{\mathsf{T}}$ *and* $M(s) = \Sigma(s)\,[I + \Sigma(s)]^{-1}$*. Then*

$$\begin{aligned} -2\log L(s,\tau) &= \|\boldsymbol{y}_{\mathrm{w}}\|^2 - \|\boldsymbol{x}\|^2 + \|\boldsymbol{x} - \boldsymbol{m}\|^2 \\ &- (\boldsymbol{x} - \boldsymbol{m})^{\mathsf{T}} Q M Q^{\mathsf{T}} (\boldsymbol{x} - \boldsymbol{m}) + \log\det(I + \Sigma) + \textit{const.} \end{aligned} \tag{11}$$

Proof. Let $U = T_\tau V^{\mathsf{T}}$, which has orthonormal columns. The Woodbury identity [40, 41] gives $(I + U\Sigma U^{\mathsf{T}})^{-1} = I - U(\Sigma^{-1} + I)^{-1} U^{\mathsf{T}} = I - UMU^{\mathsf{T}}$; the last form holds also for singular Σ. The determinant lemma gives $\det(I + U\Sigma U^{\mathsf{T}}) = \det(I + \Sigma U^{\mathsf{T}} U) = \det(I + \Sigma)$. For the residual $\boldsymbol{r} = \boldsymbol{y}_{\mathrm{w}} - T_\tau \boldsymbol{\mu}$, orthogonality of T_τ gives $\boldsymbol{r}^{\mathsf{T}}\boldsymbol{r} = \|T_{-\tau}\boldsymbol{y}_{\mathrm{w}} - \boldsymbol{\mu}\|^2$ and $U^{\mathsf{T}}\boldsymbol{r} = V(T_{-\tau}\boldsymbol{y}_{\mathrm{w}} - \boldsymbol{\mu})$. Decompose $T_{-\tau}\boldsymbol{y}_{\mathrm{w}}$ into its projection $\tilde{B}^{\mathsf{T}}\boldsymbol{x}$ and an orthogonal remainder. Since $\boldsymbol{\mu} = \tilde{B}^{\mathsf{T}}\boldsymbol{m}$ and $V = Q^{\mathsf{T}}\tilde{B}$, we get $\|T_{-\tau}\boldsymbol{y}_{\mathrm{w}} - \boldsymbol{\mu}\|^2 = \|\boldsymbol{y}_{\mathrm{w}}\|^2 - \|\boldsymbol{x}\|^2 + \|\boldsymbol{x} - \boldsymbol{m}\|^2$ and $V(T_{-\tau}\boldsymbol{y}_{\mathrm{w}} - \boldsymbol{\mu}) = Q^{\mathsf{T}}(\boldsymbol{x} - \boldsymbol{m})$. □

a. Cost. Whitening and one real FFT cost $\mathcal{O}(N \log N)$ per record. The aligned projections $\boldsymbol{x}(\tau)$ cost $\mathcal{O}(K_b N \log N)$ for all integer lags at once, or $\mathcal{O}(K_b N)$ at one continuous τ with its first two derivatives (eq. (5)). The rest of (11) costs $\mathcal{O}(K_b K_v + K_v^3)$ per (s, τ). A dense non-stationary filter instead needs an $\mathcal{O}(N^3)$ factorisation of $R(E, t_0)$ for every energy, and for every t_0 if the shift is applied to the covariance. Alternatively it can precompute inverses on a grid at $\mathcal{O}(N^2)$ memory per node. Here $K_b = K_v = K + K_n \approx 12$, against $N = 1024$.

C. Learning V and $\Sigma(s)$ and what is identifiable

Given Stage A/B estimates $(\hat{s}_j, \hat{\tau}_j)$, the aligned whitened records $\boldsymbol{z}_j = T_{-\hat{\tau}_j} \boldsymbol{y}_{\mathrm{w}}^{(j)}$ supply residuals. We take V_r as the top-K_n eigenvectors of the pooled residual covariance projected onto the complement of $\mathrm{span}(B)$. Σ is estimated at knots s_b as $V\widehat{\mathrm{Cov}}_b(\boldsymbol{z})V^{\mathsf{T}} - I$, a $K_v \times K_v$ matrix that is well determined from a few hundred events. Between knots it is interpolated through a square-root factor: the positive part is written as FF^{T}, consecutive factors are aligned by orthogonal Procrustes rotation, and F is interpolated with a monotone cubic (the small negative part entry-wise). Interpolating covariance entries instead turns a fluctuation whose direction rotates between two lines into a weaker two-direction mixture, which misdirects the oblique projection between sparse calibration lines (section VIII J). Eigenvalues are clipped

at with $\Sigma(0) = 0$, and interpolated with a monotone cubic (PCHIP) per entry, with eigenvalues clipped at $-\frac{1}{2}$. The definition of the residual matters.

Proposition 5 (Tangential fluctuations are not identifiable from continuum data)**.** *Let records at energy E fluctuate as $\boldsymbol{\mu}(s)+\xi\,\boldsymbol{d}$ with ξ zero-mean and $\boldsymbol{d} = \alpha\,\boldsymbol{\mu}'(s)+\boldsymbol{d}_\perp$, $\boldsymbol{d}_\perp \perp \boldsymbol{\mu}'(s)$. To first order in ξ, the distribution of records is the same as that of a model with fluctuation $\xi\boldsymbol{d}_\perp$ and an energy distribution $p(s)$ convolved with the law of $\alpha\xi$. If p is unknown and smooth (a continuum), the two models cannot be distinguished. If p has atoms (mono-energetic lines), the convolution is detectable and α is identified.*

Proof. $\boldsymbol{\mu}(s)+\xi\alpha\boldsymbol{\mu}'(s)+\xi\boldsymbol{d}_\perp = \boldsymbol{\mu}(s+\alpha\xi)+\xi\boldsymbol{d}_\perp+\mathcal{O}(\xi^2)$. □

Consequently, residuals about the fitted curve, $\boldsymbol{z}_j - B^\mathsf{T}\boldsymbol{c}(\hat{s}_j)$, reveal only $\boldsymbol{d}_\perp$; we call this label-free variant "curve". Since $\boldsymbol{d}_\perp \perp \boldsymbol{\mu}'$, down-weighting it does not change the energy estimate to first order. The "line-aware" variant instead takes residuals about the mean of each mono-energetic cluster, found as a peak of the histogram of $\hat{s}$. This reveals the full $\boldsymbol{d}$, including the tangential part that broadens the line. The resulting ML estimate is the oblique projection along $\boldsymbol{d}$ (cf. Ref. [8]). It is a learned, multi-dimensional generalisation of the empirical rise-time and arrival-phase corrections used in practice [4]. No energies are needed for this step, only the existence of sharp lines. Lines with a poor peak-to-continuum ratio contaminate the cluster covariance with genuine energy spread.

D. Optimisation and Cramér–Rao bounds

We maximise (11) jointly over (s, τ) with a Levenberg-damped 2×2 Newton iteration and backtracking, vectorised over records. Steps in s are capped at $0.25(|s| + 40)$, because nuisance corrections scale with the signal. Derivatives in τ are analytic, via (5); derivatives in s are central differences of the K_b-dimensional expression. The iteration starts from the Stage A/B estimates. The Fisher information of the learned model is [42]

$$I_{ij} = \partial_i\boldsymbol{\mu}^\mathsf{T}R_w^{-1}\partial_j\boldsymbol{\mu} + \tfrac{1}{2}\,\mathrm{tr}\big(R_w^{-1}\partial_i R_w\,R_w^{-1}\partial_j R_w\big)\,, \quad (12)$$

$$\partial_\tau R_w = -(DV^\mathsf{T}\Sigma V + V^\mathsf{T}\Sigma V D^\mathsf{T}), \quad (13)$$

with $\partial_s R_w = V^\mathsf{T}\Sigma'(s)V$. We report bounds with and without the covariance term, because the latter relies on the exact energy dependence of Σ.

Figure 1 summarises the procedure.

Require: pulse records $\{\boldsymbol{y}_j\}$, pulse-free records, K, K_n, number of lines
1: $\lambda_k \leftarrow$ periodogram of pulse-free records; $\boldsymbol{y}_\mathrm{w} \leftarrow W\boldsymbol{y}$ ▷ lemma 1
2: $B \leftarrow$ top-K SVD of $\{\boldsymbol{y}_\mathrm{w}^{(j)}\}$ with SNR $>$ SNR$_\mathrm{min}$
3: **repeat** ▷ Stage A
4: $\quad\tau_j \leftarrow \arg\max_\tau \|B_{1:r}T_{-\tau}\boldsymbol{y}_\mathrm{w}^{(j)}\|^2$ (FFT lags, then Newton)
5: $\quad B \leftarrow$ top-K right singular vectors of $\{T_{-\tau_j}\boldsymbol{y}_\mathrm{w}^{(j)}\}$; regauge
6: **until** J_r converged
7: $\boldsymbol{a}_j \leftarrow BT_{-\tau_j}\boldsymbol{y}_\mathrm{w}^{(j)}$; fit spline graph $\boldsymbol{c}$; $s \leftarrow$ arc length ▷ Stage B
8: $(s_j, \tau_j) \leftarrow$ stationary ML (eq. (11) with $\Sigma = 0$)
9: clusters $\leftarrow$ peaks of $\{s_j\}$; $V, \Sigma(s) \leftarrow$ cluster residual covariances ▷ Stage C
10: $(s_j, \tau_j) \leftarrow \arg\min$ eq. (11); $E_j \leftarrow E(s_j)$ from reference lines

FIG. 1. The template-free pulse processing algorithm.

VII. PHYSICAL SIMULATION

To test the method on effects that are physical rather than put in by hand, we integrate the coupled electro-thermal Langevin equations of a voltage-biased one-body TES [2]:

$$C\,\dot{T} = -K(T^n - T_b^n) + I^2R(T) + P_\mathrm{dep}(t) + \delta P_\mathrm{ph} - I\,\delta V_J, \quad (14)$$

$$L\,\dot{I} = V_b - I\,[R_L + R(T)] + \delta V_J + \delta V_L, \quad (15)$$

with $R(T) = \frac{1}{2}R_n\{1 + \tanh[(T - T_c)/w]\}$. The noise sources are white with one-sided spectral densities $4k_BTR(T)$ (TES Johnson), $4k_BT_bR_L$ (shunt Johnson) and $4k_BT^2G(T)F$ (thermal link, $G = nKT^{n-1}$). They are evaluated at the *instantaneous* T and R, so the noise is genuinely non-stationary during a pulse. A white SQUID current noise is added at the output. The energy is deposited through an absorber thermalisation profile $P_\mathrm{dep} = (E/\tau_\mathrm{th})e^{-(t-t_0)/\tau_\mathrm{th}}\Theta(t - t_0)$, integrated exactly over each time step so that arrival times are continuous. Position dependence enters through $\tau_\mathrm{th} = \tau_{\mathrm{th},0}(1 + \kappa\eta)$ with $\eta \sim \mathcal{U}(-1, 1)$. We integrate with the Euler–Maruyama method [43], using eight sub-steps per output sample and a 1 ms noise burn-in before every record. The ADC is modelled as a boxcar average over each sample period, so the residual aliasing of the pulse onset is present, as in a real digitiser. Table I lists the parameters.

a. Spectra and evaluation protocol. Each scenario has a training spectrum of 2×10^4 unlabelled events: 60% in lines at 1.5, 3.0, 4.5, 5.9, 8.0, 10.0 and 12.0 keV and 40% in a flat continuum from 0.5 to 13 keV. A statistically independent test spectrum of the same size has lines at 1.0, 2.2, 3.7, 5.0, 6.5, 7.2, 9.0 and 11.0 keV, all *between* the training lines. This tests the interpolation of the manifold and of $\Sigma(s)$. There are also 3000 noise records. All models are learned on the training set only. To compare estimators independently of calibration interpolation, we convert each estimator's line width to eV with its own local gain $d\langle\hat{x}\rangle/dE$. The gain is measured by local quadratic regression on the test continuum within ±600 eV of the line. FWHM values are $2.355\times$ the clipped (4σ) standard deviation. Timing jitter is the ro-

TABLE I. Simulation parameters and derived bias-point quantities.

T_c, T_b	100 mK, 55 mK	C, $G(T_c)$, n	1.2 pJ/K, 250 pW/K, 3
w, R_n, R_0/R_n	1.4 mK, 10 mΩ, 0.30	R_L, L	0.3 mΩ, 40 nH
F, SQUID noise	0.6, 6 pA/$\sqrt{\text{Hz}}$	$\tau_{\text{th},0}$, κ	12 µs, 0.02 (0 in scenario b)
f_s, N, pre-trigger	200 kHz, 1024, 256	arrival jitter	$\mathcal{U}(-2, 2)$ samples
α, loop gain	99, 27.5	τ_{eff}, I_0	170 µs, 47.6 µA

bust standard deviation of $\hat{t}_0 - t_0$ within each line. A common offset (time walk) is removed per line; that offset is a gauge choice (section IX). The two scenarios are (a) $\kappa = 0.02$ (a ±2% thermalisation-time spread) and (b) $\kappa = 0$.

b. Estimators compared. (1) OF with arrival-time search and a template averaged from the 5.9 keV training line. (2) The norm of the Stage A coefficient vector, in the spirit of Ref. [21]. (3) Stage B closest-point projection. (4) Joint (s, τ) ML with stationary noise. (5) Woodbury ML with label-free ("curve") covariance. (6) Woodbury ML with line-aware covariance. As references we use the CRB of the learned models and an oracle: a local generalised least squares estimator whose covariance (K_n modes) is estimated on half of the *test* line itself and applied to the other half (cross-fitted). Unless stated otherwise, $K = 4$, $r = 1$, $K_n = 8$, $\text{SNR}_{\min} = 30$ and a search window of ±8 samples.

VIII. RESULTS

A. Pulses and noise

Figure 2 shows the simulated detector. The peak current per unit energy falls by a factor of 2.0 between 1 and 12 keV, while the decay lengthens. This is the shape evolution that biases and degrades a fixed-template OF. Panel (d) shows that the noise variance during the pulse departs strongly from baseline. It falls *below* baseline while the sensor is driven up the transition: the loop gain and the current responsivity drop, so the same noise sources produce less current noise. This is mainly a change of the electro-thermal *response*, not of the sources. With the source amplitudes frozen at their bias-point values (dotted) the drop is even larger, because the hotter, more resistive sensor then no longer partly compensates with larger Johnson and phonon noise.

B. Learned subspace and manifold

Stage A converged in 3 iterations (fig. 3). The leading whitened basis vector is a sharpened mean pulse. The next ones encode the lengthening decay and the changing rise, and the fourth vector holds the high-frequency structure of the onset. Coloured back with $R_0^{1/2}$, they become the mean pulse and its shape-variation modes in current units. The coefficients trace a smooth curve whose transverse excursions grow with energy. The spline graph fits it with a GCV-selected penalty, and its arc length reaches $s_{\max} = 12389$ (the SNR at about 13 keV).

C. Structure of the pulse-correlated noise

Figure 4(a) shows the eigenvalue spectrum of the residual covariance about the cluster means. A few eigenvalues stand far above the Marchenko–Pastur edge [44] of the noise bulk, so the excess covariance is genuinely low-rank. Correlating the mode amplitudes with the simulation truth identifies the leading residual modes outside the signal subspace. The largest (pooled eigenvalue ≈256) follows the sampling phase, $\sin 2\pi t_0$, with $|r| \approx 0.95$. It is half of the aliasing pair of the pulse onset, which no Fourier-shifted template can represent; its partner lies largely inside span(B), where the signal-subspace part of Σ models it. The second (≈76) is driven by the thermalisation parameter η ($|r| = 0.96$–0.99), and the third mixes η and phase. Beyond these, the spectrum is at the Marchenko–Pastur edge. The physical non-stationary noise of fig. 2(d) appears as *reduced* variance, partly inside the signal subspace. This is why the signal subspace is part of the covariance model (section VI). Panel (c) shows its smallest eigenvalue falling below baseline with energy.

D. Energy resolution

Figure 5 and table II summarise the energy resolution on the held-out test lines.

a. Against the plain optimal filter. In scenario (a), the OF degrades from 2.0 to 12.4 eV between 1 and 11 keV, and the PCA norm behaves similarly. The stationary manifold ML removes the template-mismatch penalty and reaches 6.0 eV at 11 keV. Its resolution is then limited by the tangential part of the rise-time and sampling-phase fluctuations. As proposition 5 predicts, the label-free Woodbury variant leaves this limit unchanged. The line-aware Woodbury ML removes it and reaches 1.7–2.6 eV, a factor of 1.2× to 4.8× better than the OF. It agrees with the oracle local Gaussian model to within 0.04 eV at every test line. That oracle uses the test line's own covariance; ours is interpolated from training lines at other energies.

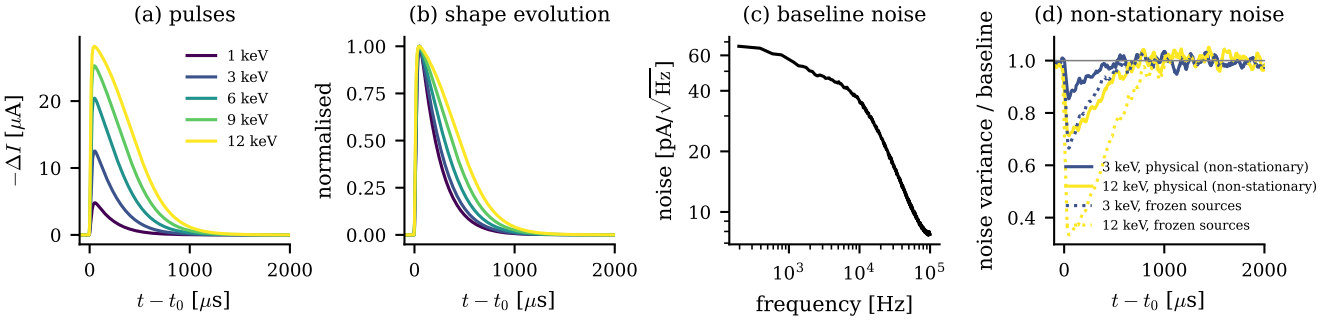


FIG. 2. Simulated TES. (a) Mean pulses at 1–12 keV. (b) The same, normalised, showing saturation and the lengthening decay. (c) Baseline noise spectrum. (d) Noise variance during the pulse relative to baseline (1500 records per curve), for the physical model (solid) and with the noise-source amplitudes frozen at the bias point (dotted).

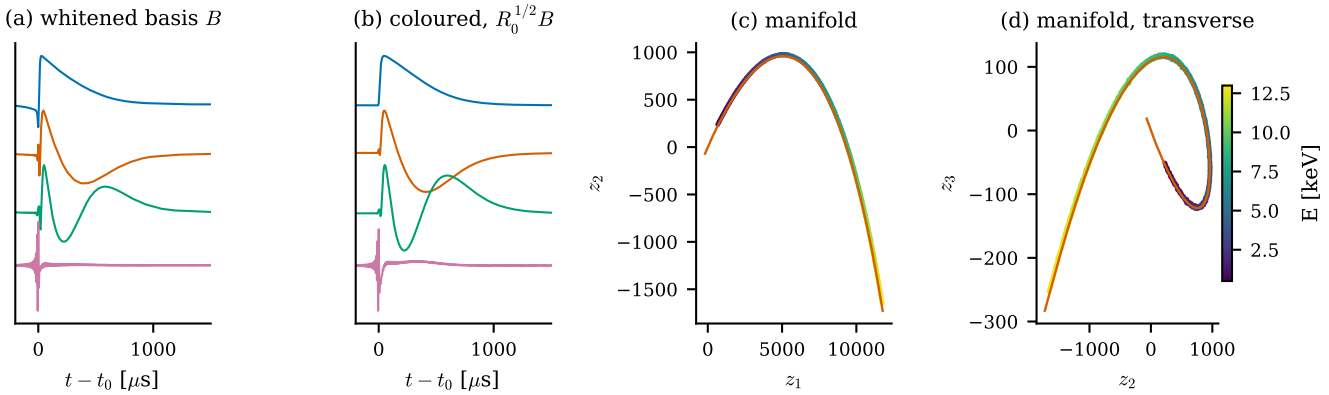


FIG. 3. (a) Whitened shift-invariant basis ($K = 4$). (b) The same vectors in the raw domain. (c, d) Aligned coefficients of the training spectrum in the rotated frame of the manifold, coloured by true energy, with the fitted curve.

b. Against prior-art pipelines. The comparison with a plain OF overstates the practical gain. The practitioner's correction chain recovers most of it: OF, phase correction, and per-line linear decorrelation against residual projectors, interpolated in energy. That chain reaches 3.1 eV at 11 keV in scenario (a). In scenario (b), phase correction alone is the best baseline (2.8 eV). Against the best of the four baselines at each line, our estimator is level at the lowest energies and better by up to 18% in scenario (a) and 8% in scenario (b). The advantage grows with saturation, where the shape fluctuations, the in-pulse noise and the energy tangent interact most. The energy-dependent full-covariance filter of Fixsen *et al.* and SIRENA is the most principled baseline. It reaches only 3.3 eV at 11 keV in (a) and 3.3 eV in (b), even with Ledoit–Wolf shrinkage toward the whitened identity. About 1700 pulses per line are too few to estimate a 1022-dimensional covariance, whereas the $K_v \approx 12$ dimensional covariance of our model is well determined. This is the practical content of the low-rank parameterisation, beyond its speed. The stationary secant filter, a discrete form of tangent filtering, tracks the stationary manifold ML, as expected for two estimates of the same curve.

c. The role of the signal subspace. A first version of our noise model used only the K_n largest residual modes. In scenario (b) it was *worse* than the phase-corrected OF at mid energies (2.14 against 2.00 eV at 6.5 keV), as was the per-line oracle in the same model class. Including the signal basis B in V (section VI) closed the gap: 2.00 eV at 6.5 keV and 2.57 eV at 11 keV, against 2.77 eV before. In this simulation the in-pulse noise along the pulse-shape directions is partly below baseline, for the electro-thermal reason of fig. 2(d). A variance-ranked residual model cannot represent this, yet it matters directly for the energy tangent. Two alternatives we tested did not help. A multiplicative in-pulse noise gain $g(t; E)$ was worse. A full-rank shrunk covariance was limited by sampling noise, as described above.

d. Cramér–Rao bounds. The CRB of the learned non-stationary model predicts the achieved widths to within about ±8% in scenario (b) and is conservative by up to 13% at high energy in (a). The stationary CRB is conservative throughout, because the in-pulse noise is partly below baseline. These are model-based predictions, not strict bounds on the simulated process. The

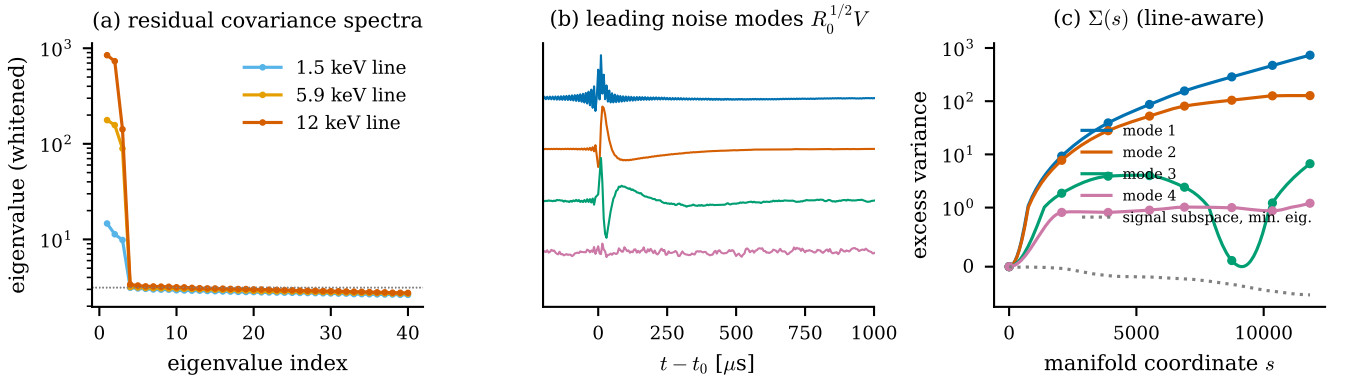


FIG. 4. (a) Eigenvalues of the whitened residual covariance of three training lines; dotted: Marchenko–Pastur upper edge for pure noise. (b) Leading noise modes in the raw domain. (c) Diagonal of the interpolated excess covariance $\Sigma(s)$ (symlog scale), with the line knots.

oracle is the practical reference.

E. Timing

Figure 7 compares the arrival-time jitter with the CRB. Without morphology variation (b), the jitter equals the CRB at 1 keV (30 ns against 29 ns). At higher energies it levels off at 18–30 ns, while the bound keeps falling to 3.2 ns. The floor is deterministic. Regressing the timing error on three harmonics of the *true* subsample phase removes it, and the remaining jitter is within a factor 1.11 of the CRB at every line (3.6 ns at 11 keV). The cause is the same residual aliasing of the pulse onset that produces the sampling-phase noise modes: a pulse that is not band-limited is displaced differently depending on where it falls between samples. The low-rank covariance treats these modes as random, which is correct for the energy estimate, but the phase-dependent displacement is a *bias* in t_0. Without labels, the bias is identifiable only through the non-uniformity of the histogram of $\hat{t}_0$ mod 1, because true arrival phases are uniform. A bias of amplitude b makes that density non-uniform by about $2\pi b$. Its Fourier estimate from n records has standard error $\sqrt{2/n}$, so a reliable correction needs $n \gg 2/(2\pi b)^2$, about 10^5 records for $b \approx 0.004$ samples. Equalising the phase histogram on our ∼1700 records per line therefore failed: the correction added more scatter than it removed. Within bins of a continuum it failed for a second reason, because the energy-dependent walk smears the histogram. With enough statistics it works. On mono-energetic sets of 10^5 simulated records, a three-harmonic `PhaseEqualizer` learned on 50 000 records reduces the jitter at 11 keV from 29.9 to 6.4 ns, against a CRB of 3.2 ns, and at 2.2 keV from 18.0 to 13.9 ns (CRB 13.5 ns). The residual gap to the bound at 11 keV reflects the three-harmonic truncation and the finite learning set. With 1000 learning records the correction is harmful (35.2 ns at 11 keV; fig. 6). Real spectra of 10^6 events per line make it a practical, label-free correction. At low statistics a phase-resolved manifold $\boldsymbol{c}(s, \varphi)$ would be needed.

With morphology variation (a), all estimators sit at 56–88 ns: a ±2% change of the thermalisation time moves the apparent arrival time by more than the noise does. Under shape variation the definition of t_0 is itself a gauge choice (section IX).

F. Low-energy behaviour

Figure 8 shows the response to events from 0 to 320 eV. Each estimator is calibrated by a quadratic through the origin fitted at 80–320 eV, because the detector is slightly non-linear even there. The coefficient norm is biased upward by 1.4 eV at zero energy, and the bias decays only over the first few tens of eV, as proposition 4 predicts. The OF with an arrival-time search is also biased (0.36 eV): maximising the filter output over lags selects positive noise excursions. The manifold ML is unbiased (0.05 eV), because its search over τ is symmetric in the sign of s. The price is a 34% wider, bimodal zero-energy distribution, since the free arrival time picks the largest $|s|$ of either sign. Above a few eV all three have the same spread.

G. Computational cost

Figure 9 compares the time per likelihood evaluation of the K_b-dimensional formula (11) with a dense Cholesky evaluation of the same non-stationary model. The two agree to a relative precision of 10^{-16} for all N. At $N = 1024$ the Woodbury evaluation is 2204 times faster per (E, t_0), and the advantage grows with N. The full pipeline processes 467 records/s on one CPU core for the line-aware ML, against 2407 for the OF, in our pure-NumPy implementation.

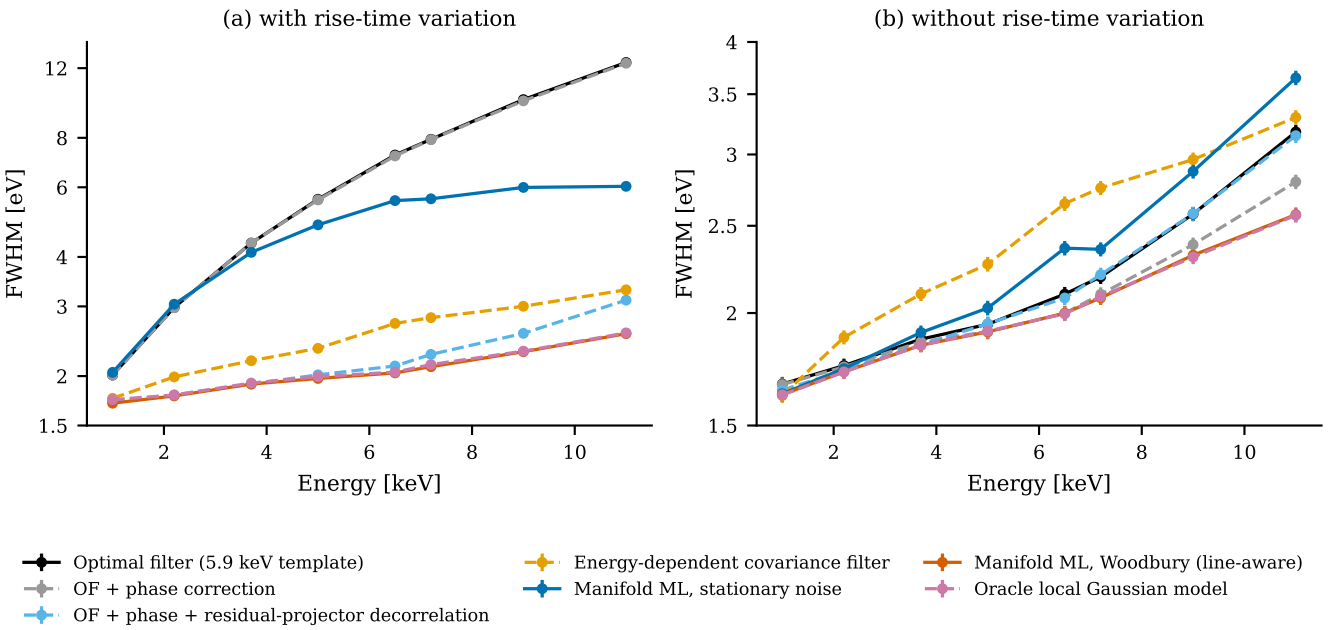


FIG. 5. Energy resolution (FWHM) on held-out test lines that lie between the training lines. (a) With ±2% thermalisation-time spread; (b) without. Solid: plain OF and the methods of this paper. Dashed: prior-art pipelines (phase-corrected OF, OF with residual-projector decorrelation, energy-dependent full-covariance filter) and the oracle local Gaussian model. Error bars are statistical. The plain and phase-corrected OF coincide in (a).

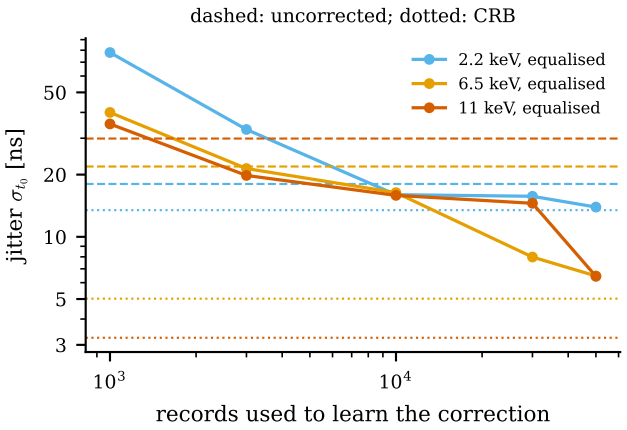


FIG. 6. Arrival-time jitter after label-free phase-histogram equalisation, versus the number of records used to learn it (mono-energetic simulated sets, no morphology variation). Dashed: uncorrected. Dotted: CRB.

H. Ablations

Figure 10 varies K and K_n. The resolution is insensitive to K between 2 and 8: the manifold is smooth enough that two whitened components describe it, and extra components neither help nor hurt. With the signal subspace in the covariance, a single residual mode (the sampling-phase mode) is not enough. From $K_n = 2$, the rise-time mode is included, the resolution reaches the oracle value, and adding more modes neither helps nor hurts noticeably. Before the signal subspace was included, ranking residual modes by the Gaussian likelihood gain $\lambda - 1 - \log\lambda$ instead of by variance helped by 1–4% at the highest energies, because it kept some directions of reduced noise. The signal-subspace covariance captures this physics more directly, and we keep variance ranking for the residual modes.

I. Real TES data

We applied the same code to real data distributed with NIST's MASS software [11]. The pulse records come from a TES at Advanced Photon Source beamline 29 (2015), sampled every 5.12 µs with 1024 samples, and are published there as test data for arrival-phase correction. The spectrum is dominated by one line (the source file name suggests 600 eV), with a sparse low-energy tail. A separate file holds 288 pulse-free records of the same channel. Alternate records form training and test halves, so all models are learned on 1621 line records and evaluated on 1445 others. With essentially one line, the absolute energy scale is unknown. Each estimator's width is therefore expressed as a fraction of the line energy, using its local gain relative to the OF amplitude measured on the off-line tail. Corrections fitted at the line keep the OF gain there.

With a single cluster, the manifold is the secant (sec-

TABLE II. FWHM energy resolution [eV] on the held-out test lines.

Estimator	E [keV]	1.0	2.2	3.7	5.0	6.5	7.2	9.0	11.0
Scenario with rise-time variation									
Optimal filter (5.9 keV template)		2.01	2.98	4.34	5.60	7.24	7.93	9.99	12.40
OF + phase correction		2.01	2.98	4.34	5.57	7.20	7.92	9.92	12.35
OF + phase + residual-projector decorrelation		1.73	1.79	1.91	2.01	2.12	2.27	2.56	3.11
Secant (tangent) filter, stationary		2.04	3.05	4.12	4.78	5.34	5.73	6.03	6.14
Energy-dependent covariance filter		1.76	1.99	2.19	2.35	2.72	2.81	3.00	3.30
PCA coefficient norm		2.06	3.18	4.57	5.73	7.10	7.59	8.96	10.13
Manifold projection (Stage B)		2.04	3.04	4.11	4.82	5.55	5.61	5.99	6.03
Manifold ML, stationary noise		2.04	3.04	4.11	4.82	5.55	5.61	5.99	6.03
Manifold ML, Woodbury (label-free)		2.04	3.01	4.09	4.81	5.49	5.62	6.03	6.07
Manifold ML, Woodbury (line-aware)		1.71	1.78	1.91	1.97	2.04	2.11	2.30	2.56
Oracle local Gaussian model		1.74	1.79	1.92	1.99	2.05	2.14	2.31	2.57
CRB, learned model		1.73	1.88	2.03	2.22	2.57	2.65	2.76	2.89
Scenario without rise-time variation									
Optimal filter (5.9 keV template)		1.67	1.75	1.87	1.94	2.10	2.19	2.58	3.18
OF + phase correction		1.67	1.74	1.86	1.91	2.00	2.10	2.38	2.80
OF + phase + residual-projector decorrelation		1.64	1.73	1.84	1.95	2.08	2.21	2.58	3.15
Secant (tangent) filter, stationary		1.62	1.74	1.90	2.04	2.39	2.36	2.92	3.69
Energy-dependent covariance filter		1.62	1.88	2.10	2.27	2.65	2.75	2.96	3.30
PCA coefficient norm		1.62	1.72	1.86	1.94	2.11	2.18	2.56	3.13
Manifold projection (Stage B)		1.63	1.74	1.90	2.03	2.36	2.35	2.87	3.65
Manifold ML, stationary noise		1.63	1.74	1.90	2.03	2.36	2.35	2.87	3.65
Manifold ML, Woodbury (label-free)		1.63	1.74	1.90	2.03	2.35	2.38	2.92	3.71
Manifold ML, Woodbury (line-aware)		1.62	1.72	1.84	1.91	2.00	2.08	2.32	2.57
Oracle local Gaussian model		1.62	1.72	1.84	1.91	2.00	2.09	2.31	2.57
CRB, learned model		1.66	1.72	1.76	1.90	2.14	2.17	2.40	2.75

tion V), and the stationary ML coincides with the OF (0.458% against 0.460% FWHM). Our reimplementations of the empirical corrections reach 0.309% (phase correction) and 0.290% (adding robust residual-projector decorrelation). The line-aware Woodbury ML reaches 0.276±0.006%. The decisive comparison is against MASS itself, run with its documented standard pipeline: summary and cuts, noise, average pulse, filter, drift correction against the pre-trigger mean, and entropy-minimising phase correction. MASS fits these corrections on all records, so its numbers are in-sample, which can only favour it. With its arrival-time-safe (ATS) filter, MASS reaches 0.254±0.005%; with the 5-lag filter, 0.265%. MASS's drift correction uses the pre-trigger level, which our whitened processing discards. Adding the same information to our estimator as a robust, line-learned gain covariate gives 0.263±0.005%. That is level with the 5-lag MASS pipeline and 1.2σ behind the ATS pipeline, against a baseline-noise limit of 0.217% (fig. 11).

On these real single-line data, then, the method matches but does not beat the best practice, and MASS's real pipeline outperforms our reimplementation of the same corrections. The simulated baselines of section VIII were built the same way and may understate real practice. We tried to run MASS on the simulated records too, but with default settings it did not reproduce even the plain OF resolution, a configuration mismatch we did not pursue. Two further lessons came from these data. First, the curve must not be fitted to the scatter inside a line. Before the cluster-mean rule was introduced, the spline followed the phase-driven scatter and the manifold estimator was four times worse than the OF. Second, projector regressions must be robust to pile-up. This single-line set does not test the non-linear energy manifold itself.

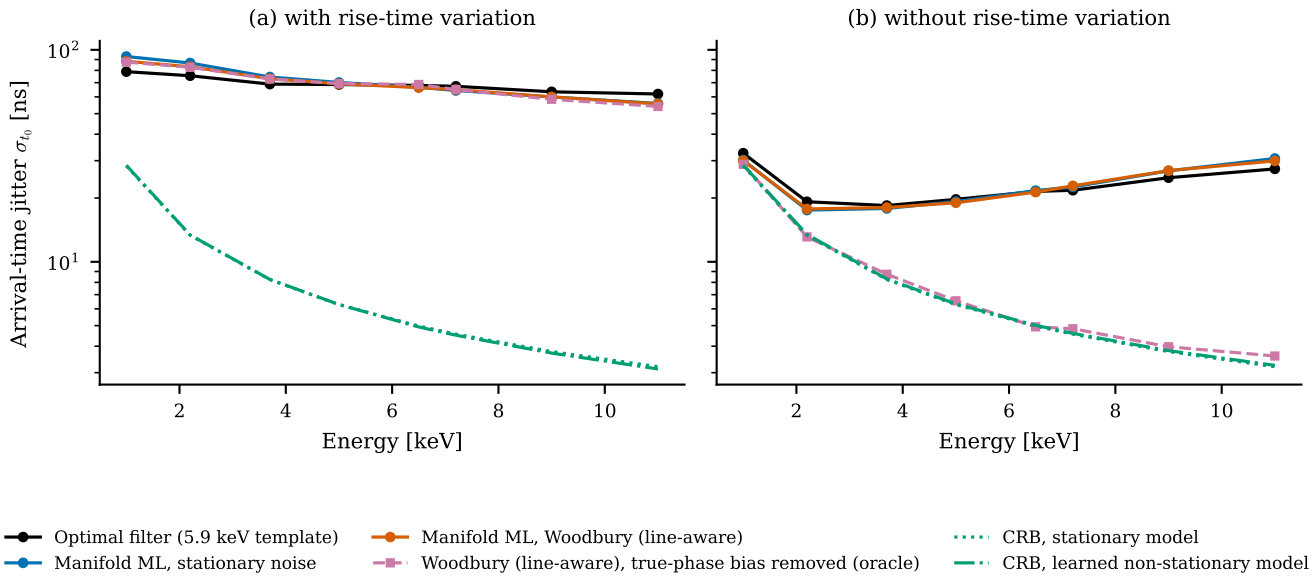


FIG. 7. Arrival-time jitter per test line (per-line time-walk offset removed; 1 sample = 5 µs). Dashed: line-aware ML after regressing out the dependence on the true sub-sample phase (oracle diagnostic). Dotted and dash-dotted: CRB of the learned stationary and non-stationary models.

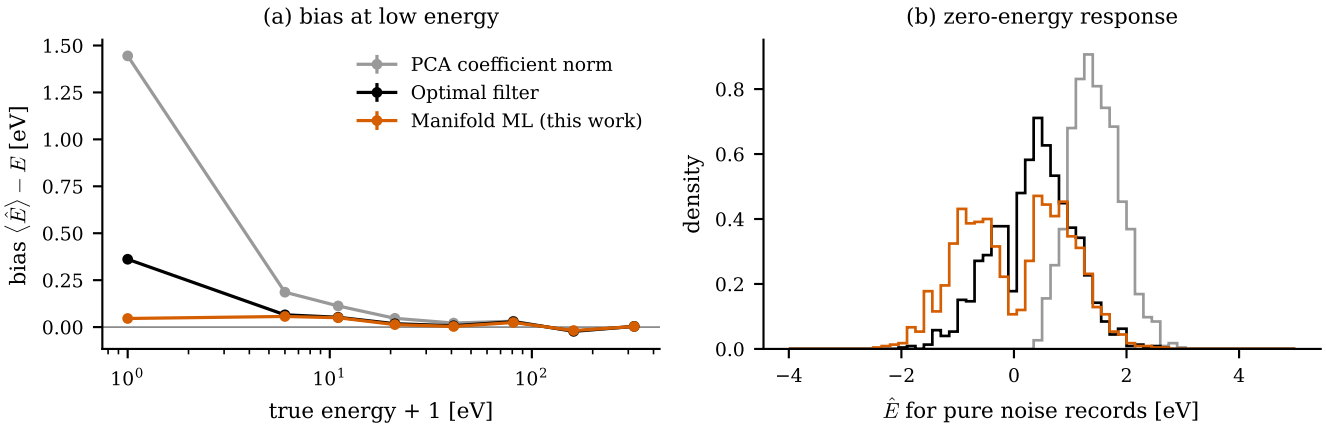


FIG. 8. (a) Bias of three estimators at low energy. (b) Distribution of the estimates for pure-noise records.

J. Phonon-mediated detector with line-free science data

Dark-matter detectors read athermal phonons from a large crystal. Their science data contain no lines, and their pulse shapes depend strongly on the event position. We extend the simulator with a crystal body (two-component pulses: fast athermal collection plus slow crystal heating, in the spirit of the CRESST pulse model) and with uniformly distributed 3-D event positions. A path-length proxy $d = (\rho^2 + z)/2$ sets the athermal collection efficiency, $\varepsilon \propto 1 - 0.2d$, and time, $\tau_a = \tau_{a,0}(1+d)$, so amplitude and shape change together. A further 1% amplitude-only radial term cannot be separated from energy by any single-channel estimator. A calibration run (2×10^4 events; 70% in lines at 1.3, 5.9, 10.4, 14.4, 17.5 and 22.2 keV on a flat continuum) and an independent science run without lines (2×10^4 events, flat plus exponential background, 0.5–25 keV) are simulated. All energy scales come from the calibration lines, whose identities are given to every line-aware method, ours and the baselines alike. The resolution is scored on the science run from the simulation truth: the FWHM of the energy error in bins of true energy, after removing a local linear trend so that calibration-interpolation bias is not counted.

Figure 12 shows the outcome. Without lines, the label-free variant trained on science data cannot remove the

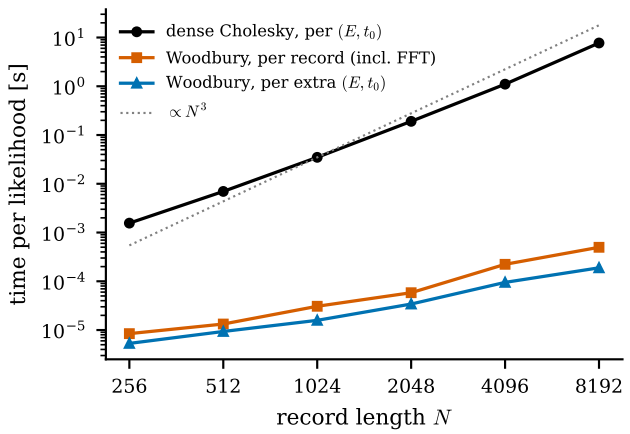


FIG. 9. Time per evaluation of the non-stationary log-likelihood: dense Cholesky versus theorem 1 ($K_s = 4$, $K_n = 8$).

position dependence (712 eV at 5 keV), and neither can the OF (722 eV). This is proposition 5 in its starkest form. With the covariance learned on the calibration run, our estimator reaches 21, 33, 65 and 126 eV at 1, 5, 13.75 and 21.25 keV. That is 1.2–3.6 times better than the energy-dependent full-covariance filter trained on the same calibration run (74, 52, 124 and 253 eV), across the whole range covered by calibration lines. Our implementation of the empirical projector correction works at low energy (22 eV at 1 keV) but fails at high energy, where position effects are large and non-linear. Given the real-data experience, a better-tuned version may do better. Above the last calibration line our estimator extrapolates and loses its advantage (401 against 268 eV at 23.75 keV). Adding the science run to the training, so that its continuum constrains the manifold, changes little (33 eV at 5 keV). Two implementation details were essential here. First, the step size of the likelihood maximisation must scale with the signal. Second, $\Sigma(s)$ must be interpolated through aligned square-root factors: entry-wise interpolation of Σ gave errors of several hundred eV between the sparse calibration lines.

IX. DISCUSSION

a. What the simulations support. Against a plain fixed-template OF, the template-free, line-aware estimator improves the FWHM by 1.2× to 4.8× on held-out lines of a saturating TES with a ±2% thermalisation spread. Against the best of four prior-art pipelines it is level at low energy and up to 18% better at the highest energies. It matches a per-line oracle Gaussian model to within 0.04 eV and needs far fewer calibration pulses than an energy-dependent full-covariance filter. The low-energy biases of the coefficient norm and of the OF with time search are removed. Without morphology variation, timing reaches the CRB once a deterministic sampling-phase bias is removed. The non-stationary likelihood costs about as much as a stationary filter with K_b templates. The method's value is therefore less a large resolution gain over the best practice than the fact that it reaches and slightly exceeds that practice from a single, template-free likelihood. That likelihood also yields arrival times, Cramér–Rao predictions and a goodness of fit, without hand-built correction stages.

b. What needs qualification. Several statements of the original research plan have to be weakened. First, "calibration independence" holds for the pulse manifold and for the noise model, which are learned from unlabelled data. The absolute energy scale $E(s)$ still needs reference lines, and the tangential part of the covariance needs mono-energetic clusters (proposition 5). Second, Cholesky pre-whitening must be replaced by the symmetric spectral root for shift invariance to be exact (lemma 1). Third, the cost of the non-stationary likelihood is $\mathcal{O}(N \log N)$ per record for the transform, plus $\mathcal{O}(K_b N)$ per continuous (E, t_0) evaluation (or $\mathcal{O}(K_b K_n)$ once the projections are tabulated). It is not $\mathcal{O}(N \log N)$ for the inverse itself, which is never formed. Fourth, the CRB of the learned low-rank model is a model-based prediction, accurate to roughly 10% here, not a strict bound on the physical process. Fifth, the "$> 2\times$ FWHM improvement in saturated regimes" holds only against a plain OF and only when shape fluctuations such as position dependence or sampling-phase effects are present. Against phase correction and projector decorrelation, the gain is at most 18%. For a detector whose pulses form a strict one-parameter family, a fixed template with a non-linear gain curve loses at most 24% in our simulation. Sixth, sub-sample timing "locked to the Fisher limit" needs a correction of the sampling-phase bias. Phase-histogram equalisation achieves it without labels, but only with of order 10^5 records per energy.

c. Timing gauge. With energy-dependent shapes, "arrival time" is defined only up to an energy-dependent offset $\delta(E)$ (time walk): the reparameterisation $\boldsymbol{\mu}(s) \mapsto T_{\delta(s)}\boldsymbol{\mu}(s)$, $t_0 \mapsto t_0 - \delta(s)$ leaves the likelihood unchanged. Our gauge aligns the mean-pulse direction. A Fisher-orthogonal gauge, in which $\langle \boldsymbol{\mu}', D\boldsymbol{\mu} \rangle = 0$, would decorrelate energy and time estimates. A physical onset gauge needs an external timing reference. Because the noise-free manifold is known, conversion between gauges is a deterministic post-processing step.

d. Limitations and outlook. The non-linear regime is tested only in simulation, including the phonon-mediated case of section VIII J, where calibration-run covariances transfer to line-free science data but cannot be replaced by anything learned from the science data alone. The real-data test of section VIII I covers a single line and confirms the shift-invariant learning, the line-aware covariance and its robustness. Next, the method needs multi-line or continuum data from saturating arrays with pile-up, gain drift and non-Gaussian tails in the shape fluctuations. Pile-up rejection can use the likeli-

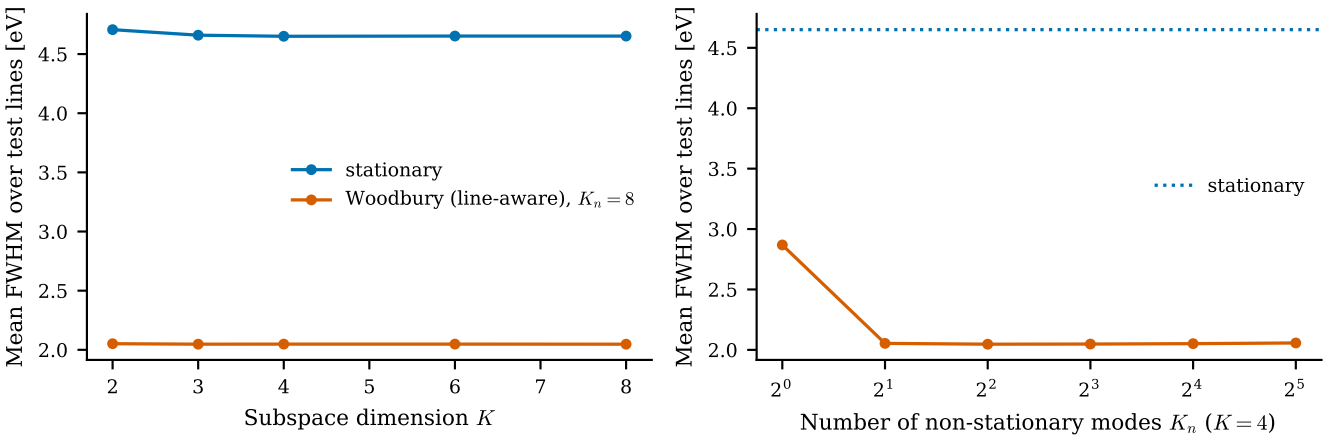


FIG. 10. Mean FWHM over the test lines versus the subspace dimension K (left) and the number of non-stationary modes K_n (right), scenario (a).

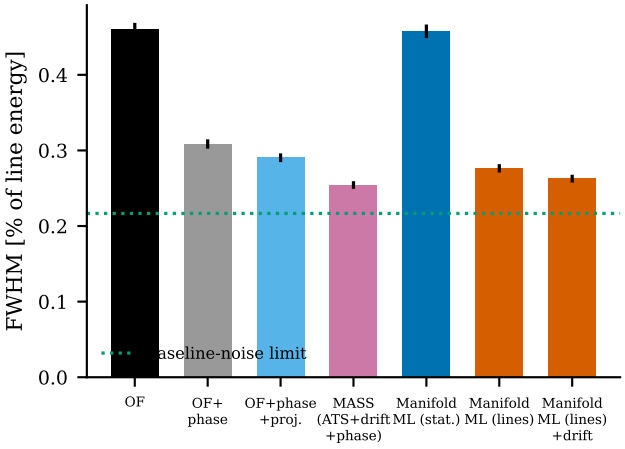


FIG. 11. Line width on held-out records of real NIST TES data, as a fraction of the line energy. MASS: NIST's standard pipeline, corrections fitted in-sample. Dotted: baseline-noise limit of the OF.

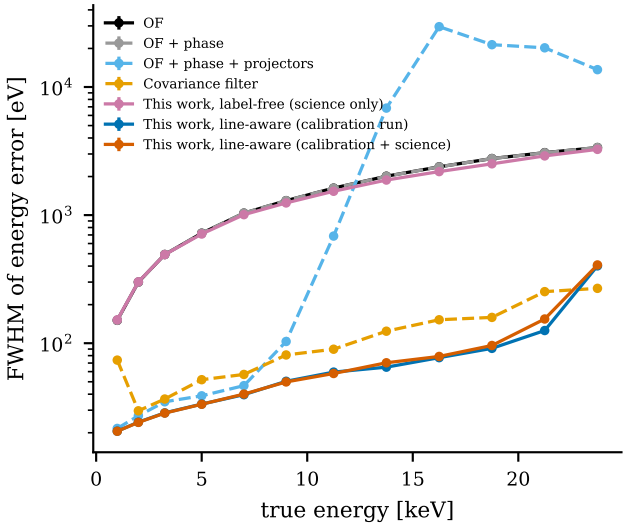


FIG. 12. Phonon-mediated detector: FWHM of the energy error on the line-free science run versus true energy. Line-aware methods use the calibration run.

hood value (11) directly. Gain drift enters as a slow reparameterisation of s and could be tracked by line centroids. Continuous morphology parameters could be estimated explicitly with a two-dimensional manifold $\boldsymbol{c}(s, \eta)$; we did not pursue this because the tangential identifiability problem applies there too, and the line-aware covariance already reaches the oracle. The low-rank model would fail for noise whose non-stationarity is spread over many modes. A diagonal-plus-low-rank model in the time domain would keep part of the speed-up.

X. CONCLUSION

We have presented a template-free pulse processing chain for non-linear microcalorimeters. Its three stages are shift-invariant PCA in the whitened domain, an arc-length spline manifold, and a Woodbury-accelerated, low-rank non-stationary maximum likelihood. Each stage optimises a stated likelihood, and each builds on an identifiable line of prior work (section II). On held-out lines of a physically simulated, saturating TES, the chain reaches the resolution of an oracle Gaussian model. It is level with the best empirical correction chain at low energy and up to 18% better at high energy, at the cost of a small bank of templates and with far fewer calibration pulses than a full-covariance filter. The open-source implementation is intended to make these methods easy to test on real detector data. A first test on single-line NIST TES data shows it level with, not ahead of, NIST's

MASS pipeline. Multi-line data from saturating arrays are the necessary next step.

DATA AND CODE AVAILABILITY

All code, including the simulator and the scripts that produce every figure and number in this paper, is available at `https://github.com/ytcs/empca`.

Appendix A: Software

The package `empca` (import name) implements every component as a separate, tested class: `StationaryNoise`, `ShiftInvariantPCA`, `PulseManifold`, `LowRankNoise`, `ManifoldFitter` and `OptimalFilter`. A pipeline object chains them:

```
from empca import PulseProcessor
proc = PulseProcessor(n_components=4, n_lines=7,
                      n_noise_modes=8)
proc.fit(train_records, noise_records)
res = proc.transform(records)  # res.s, res.t0
proc.calibrate(proc.line_centers_, line_eV)
energy = proc.energy(res.s)
fwhm = proc.resolution(res.s)  # learned-model CRB
```

`PhaseEqualizer` implements the label-free sampling-phase correction of section VIII. The frequency-domain EM-PCA of Ref. [21] remains available as `empca.EMPCA`.

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